OPTIMAL MATURITY STRUCTURE

Fiscal and Monetary Policy 2026

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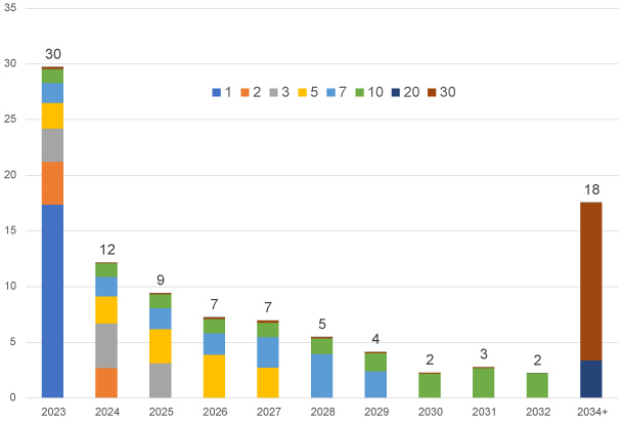
April 20, 2026

PLAN

- Why debt maturity can matter even though our earlier models emphasized one-period debt.
- Angeletos (2002): step by step, how an inherited maturity portfolio can replicate the state-by-state debt burden delivered by contingent debt.
- Angeletos' two-state example: why the hedge is short assets + long debt.
- Buera–Nicolini (2004): why exact spanning is quantitatively fragile.
- Brief modern view: Bhandari et al. (2017) and Aparisi de Lannoy et al. (2025).

MATURITY STRUCTURE IN THE U.S.

Figure 1



NOTE: We plot the fixed-rate marketable US debt (FRMUSD) maturing in each year. The colors represent the debt's maturity in years. E.g., The 1-year debt in blue contains all notes and bills with maturity of 1 year or less.

SOURCE: US Treasury quantities and rates in December 2022 are from the Monthly Statement of Public Debt (MSPD) published by the US Treasury Department (available on fiscaldata.treasury.gov/).

Source: St. Louis Fed Economic Synopses (2023)

THREE LESSONS FROM THE LITERATURE

- Lucas and Stokey (1983): with full commitment and state-contingent debt, maturity is irrelevant for allocations; contingent payoffs do the insurance job directly.
- Lucas and Stokey also note that maturity can matter for time consistency, but that is not today's focus.
- Angeletos (2002) and Buera–Nicolini (2004): if the government can trade many maturities of noncontingent real debt, the market value of its inherited portfolio can move across states.
- Modern work then asks a different quantitative question: once we discipline returns with asset-pricing data, what maturity structure best reduces fiscal risk?

FROM MATURITIES TO SPANNING

SIGN CONVENTION

- Throughout the deck, $B > 0$ means the government has issued debt: it is a liability.
- $B < 0$ means the government holds the claim as an asset.
- The same convention applies to Arrow debt:

$$b\left(s^t\right) > 0 \text{ means the government owes resources at } t.$$

- Therefore a positive market value $V_t^M\left(s^t\right)$ is a debt burden, while a negative value is a net asset position.

MECHANISM: PRICING DIFFERENT MATURITIES

- A real zero-coupon bond issued at t and maturing at $t + j$ has price

$$p_t^j\left(s^t\right) = \beta^j \mathbb{E}_t\left(\frac{u_c\left(s^{t+j}\right)}{u_c\left(s^t\right)}\right).$$

- If the bond was purchased at $t - 1$, its realized gross return at t is

$$R_t^j\left(s^t\right) = \frac{p_t^{j-1}\left(s^t\right)}{p_{t-1}^j\left(s^{t-1}\right)}, \qquad j \geq 2,$$

  while a one-period bill simply pays one unit for sure at t.

- Therefore a bill due at t and a bond with remaining maturity are different assets: one pays a fixed amount at t, the other is marked to market at t.

MECHANISM: WHY MATURITY CREATES STATE CONTINGENCY

- A bill maturing at t is worth exactly 1 in every state.
- A longer bond is worth its resale price $p_t^{j-1}\left(s^t\right)$, and that resale price depends on the realized state s_t.
- Since $p_t^{j-1}\left(s^t\right)$ moves with the state, a portfolio chosen at $t-1$ delivers different date-t market values across states.
- So maturities mimic contingent payoffs through state-dependent valuation, not through state-dependent promised cash flows.

VALUE OF AN INHERITED MATURITY PORTFOLIO

- At the beginning of date t, after s_t is realized, the government inherits

$$B_{t-1}\left(s^{t-1}\right) = \begin{bmatrix} B_{t-1}^{1}\left(s^{t-1}\right) \\ \vdots \\ B_{t-1}^{J}\left(s^{t-1}\right) \end{bmatrix}.$$

- Its date-t market value is

$$V_t^M\left(s^t\right) = B_{t-1}^{1}\left(s^{t-1}\right) + \sum_{j=2}^{J} p_t^{j-1}\left(s^t\right) B_{t-1}^{j}\left(s^{t-1}\right).$$

- The first term is noncontingent at t; the terms with $j \geq 2$ move with today's bond prices.
- Therefore the same promised payment schedule chosen at $t-1$ delivers different mark-to-market values across states at t.

WHY ONE-PERIOD DEBT ALONE CANNOT SPAN

- If the government can issue only one-period bills, then $J = 1$ and

$$V_t^M\left(s^t\right) = B_{t-1}^1\left(s^{t-1}\right)$$

  is the same in every current state.
- Then inherited debt cannot offset state-by-state variation in g_t or in the tax distortions implied by Ramsey policy.
- Long maturities matter because they leave behind claims whose resale price depends on s_t.
- With enough maturities whose prices move differently across states, the government can engineer a state-contingent effective debt burden even with noncontingent promised payoffs.

STEP 1: START FROM THE COMPLETE-MARKETS RAMSEY PLAN

- Take the complete-markets Ramsey allocation as the target.
- Keep the notation from the previous Lucas–Stokey slides:

$$s_t \text{ current state,} \qquad s^t \text{ history,} \qquad q\left(s^{t+1} \mid s^t\right) \text{ Arrow price.}$$

- In the complete-markets deck, the Ramsey plan is supported by an Arrow payoff menu $\left\{b\left(s^{t+1}\right)\right\}_{s_{t+1}}$.
- In those slides, $b\left(s^t\right)$ is the inherited Arrow obligation due today.

STEP 1: WHAT IS THE TARGET OBJECT?

- In those slides, $b\left(s^t\right)$ is the inherited Arrow obligation due today, while $\sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$ are the proceeds from selling the next Arrow menu.
- Under the complete-markets Ramsey plan, this inherited obligation equals the node-specific present value of future primary surpluses; denote it by $PV_t^A\left(s^t\right) \equiv b\left(s^t\right)$.
- Sign convention: $PV_t^A\left(s^t\right) > 0$ means the government owes that amount at date t.
- Angeletos asks: can an inherited portfolio of noncontingent maturities generate exactly this same state-by-state effective debt burden at date t?

STEP 2: ASK WHAT THE INHERITED MATURITY PORTFOLIO DELIVERS

- At date t, after the state is realized, the maturity portfolio has market value

$$V_t^M\left(s^t\right) = B_{t-1}^1\left(s^{t-1}\right) + \sum_{j=2}^{J} p_t^{j-1}\left(s^t\right) B_{t-1}^j\left(s^{t-1}\right).$$

- Sign convention: $V_t^M\left(s^t\right) > 0$ means the government owes that market value at date t.
- The quantities B_{t-1}^j were chosen one period earlier; they are not state contingent at t.
- But their value at t is state contingent, because the resale prices $p_t^{j-1}\left(s^t\right)$ vary with s_t.
- So maturities matter through valuation, not through explicitly contingent promised payments.

STEP 3: MATCH DELIVERED VALUE TO REQUIRED VALUE

- To reproduce the complete-markets allocation with maturities only, require

$$V_t^M\left(s^t\right) = PV_t^A\left(s^t\right)$$

  in every current state s_t.

- This is the whole logic of the replication:

$$\text{choose promised quantities at } t-1 \quad \Longrightarrow \quad \text{get the right state-by-state values at } t.$$

- If this equality holds for each state, the maturity portfolio delivers exactly the same date-t effective debt burden as the complete-markets Ramsey plan.

STEP 4: STACK ACROSS STATES AND SOLVE A LINEAR SYSTEM

- Fix last period's history s^{t-1} and let current states be $s_t \in \{s_1, \ldots, s_S\}$.
- Stack the equations $V_t^M\left(s^{t-1}, s_k\right) = PV_t^A\left(s^{t-1}, s_k\right)$ across current states:

STEP 4: MATRIX FORM

$$\underbrace{\begin{bmatrix} PV_t^A\left(s^{t-1}, s_1\right) \\ \vdots \\ PV_t^A\left(s^{t-1}, s_S\right) \end{bmatrix}}_{PV_t^A} = \underbrace{\begin{bmatrix} 1 & p_t^1\left(s^{t-1}, s_1\right) & \cdots & p_t^{J-1}\left(s^{t-1}, s_1\right) \\ \vdots & \vdots & & \vdots \\ 1 & p_t^1\left(s^{t-1}, s_S\right) & \cdots & p_t^{J-1}\left(s^{t-1}, s_S\right) \end{bmatrix}}_{A_t} \underbrace{\begin{bmatrix} B_{t-1}^1 \\ \vdots \\ B_{t-1}^J \end{bmatrix}}_{B_{t-1}}.$$

- Column j of A_t is the vector of date-t values across states of holding maturity j from $t-1$ to t.
- The portfolio problem is now pure linear algebra: can the target surplus-value vector PV_t^A be written as a combination of those maturity-value columns?

STEP 5: WHEN DOES THE SYSTEM HAVE A SOLUTION?

- Angeletos (2002): generically, almost every Arrow–Debreu allocation can be implemented with noncontingent debt of different maturities.
- Equivalently, find B_{t-1} such that $PV_t^A = A_t B_{t-1}$.
- If $J \geq S$ and A_t has full rank, the inherited portfolio can reproduce the required debt burden exactly.
- If $J < S$, or if the columns of A_t are linearly dependent, generic complete-markets payoffs cannot be replicated exactly.
- Economic content:
- enough maturities plus enough state variation in their prices implies the complete-markets allocation can be spanned.

TWO-STATE ANGELETOS EXAMPLE: SOLVE THE PORTFOLIO

- Two states: L and H, with $g_H > g_L$, and two maturities: a short bill and a long bond.
- After subtracting the common one-period component, normalize the target present values of surpluses so that $PV_H^A = 0$ and $PV_L^A > 0$.
- The spanning system is

$$\begin{bmatrix} PV_H^A \\ PV_L^A \end{bmatrix} = \begin{bmatrix} 1 & R_H^{-1} \\ 1 & R_L^{-1} \end{bmatrix} \begin{bmatrix} B_{t-1}^1 \\ B_{t-1}^2 \end{bmatrix},$$

  where R_H and R_L are state-dependent short real returns and $R_H > R_L$ in the bad state.

TWO-STATE ANGELETOS EXAMPLE: SIGN PATTERN

- Solve the two equations row by row:
- Solving row by row gives

$$B^2_{t-1} = \frac{PV^A_L}{R_L^{-1} - R_H^{-1}} > 0, \qquad B^1_{t-1} = -R_H^{-1} B^2_{t-1} < 0.$$

- Hence $B^1_{t-1} < 0 < B^2_{t-1}$: hold short assets, issue long debt.

TWO-STATE ANGELETOS EXAMPLE: ECONOMIC INTUITION

- In the high-spending bad state H, marginal utility is high, so the short real rate is high and the resale price of long debt is low:

$$R_H > R_L \qquad \Longleftrightarrow \qquad R_H^{-1} < R_L^{-1}.$$

- If the government has issued long debt, the market value of that liability is lower in the bad state. That is the hedge.
- If it also holds short assets, it has a reserve that can be drawn on when expenditures are high.

TWO-STATE ANGELETOS EXAMPLE: RESERVE-FUND INTUITION

- Angeletos' continuous-maturity version is sell a perpetuity and invest in a reserve fund.
- Good state: roll over the reserve fund.
- Bad state: draw it down while higher returns lower the market value of long liabilities.

QUANTITATIVE REALITY CHECK

BUERA AND NICOLINI (2004): EXACT SPANNING CAN BE EXTREME

Table 2
Debt positions and interest rates for the four states examples

Risk aversion	Debt positions $\left(\frac{p^j(g_{low},s_{high})b^j}{y_{g_{low},s_{high}}}\right)$				Short term interest rates (%)			
					g_{low}, s_{high}	g_{low}, s_{low}	g_{high}, s_{high}	g_{high}, s_{low}
Business cycle examples								
Our calibration	$j = 0.25$	$j = 1$	$j = 2$	$j = 30$				
0.50	−21.83	146.08	322.70	198.46	1.03	3.93	1.15	4.06
2.00	−10.10	65.61	−140.41	84.92	−1.15	7.68	−0.66	8.21
10.00	−9.13	60.22	−129.22	78.13	−0.44	6.16	0.07	6.73
Calibration of Chari et al. (1995)	$j = 1$	$j = 4$	$j = 13$	$j = 30$				
1.00	−24.62	71.96	−171.89	125.12	1.81	2.04	1.99	2.23
2.00	−13.58	45.01	−125.93	95.07	0.93	2.08	1.90	3.12
9.00	−7.49	20.52	−43.39	30.00	0.41	2.58	1.25	3.54

Note: In columns 2–5, we report the maturities implementing the Ramsey allocation that minimized the average value of the absolute value of the positions. The business cycle examples were calibrated to match the US postwar experience. Our calibration was done taking a quarter as the time period, while Chari et al. (1995) use the year. In our (Chari et al. (1995)) calibration the short-term interest rate corresponds to a 3-month (1-year) bond. In both cases, interest rates correspond to yearly returns.

BUERA AND NICOLINI (2004): QUANTITATIVE TAKEAWAY

- Table 2 in Buera and Nicolini (2004): even in a simple four-state calibration, implementing Ramsey can require gross positions on the order of a few hundred times GDP.
- The problem is not the sign pattern; it is the magnitude required to make the spanning equations hold exactly.

WHY DO THE POSITIONS EXPLODE?

- Spanning requires inverting the return matrix A.
- If returns at different maturities are highly correlated, A is nearly singular and the required hedging positions become huge.
- Persistence is especially problematic: in these no-capital models, more persistent shocks imply smaller cross-state movements in interest rates.
- Angeletos shows that, as persistence approaches a unit root, the short-asset / long-debt positions can explode.
- Buera and Nicolini show that the problem worsens with more states and with small changes in parameters.

WHAT THE SIMPLE NEOCLASSICAL STORY MISSES

- Angeletos already emphasizes that exact hedging requires the government to know very precisely how equilibrium interest rates move with shocks.
- If asset prices contain noise or the model of returns is misspecified, exact replication becomes unreliable.
- BEGS (2019): when we actually measure bond returns in the data, a substantial component is orthogonal to fundamentals.
- Then large maturity bets are not a free hedge: they inject additional portfolio risk into the government's budget.
- This is why the Angeletos / Buera–Nicolini portfolio is best viewed as a theoretical benchmark, not a literal quantitative prescription.

BEGS (2019): TABLE I, BOND RETURN MOMENTS

Table I: Summary Statistics for Excess Holding Period Returns

	mean	std.	p25	p50	p75	corr $\Delta \log Y$	corr $\Delta \log S$
0-6m	0.08	0.52	-0.19	0.03	0.26	-0.33	-0.22
6-12m	0.15	1.21	-0.43	0.06	0.55	-0.34	-0.23
12-18m	0.22	1.72	-0.68	0.1	0.86	-0.34	-0.23
18-24m	0.24	2.05	-0.85	0.05	1.02	-0.34	-0.25
24-30m	0.28	2.38	-0.98	0.04	1.21	-0.33	-0.24
30-36m	0.32	2.62	-1.17	0.06	1.49	-0.33	-0.25
36-42m	0.36	2.81	-1.35	0.09	1.75	-0.32	-0.25
42-48m	0.37	3.02	-1.5	-0.02	1.83	-0.33	-0.23
48-54m	0.4	3.18	-1.5	-0.12	2.02	-0.33	-0.25
54-60m	0.33	3.46	-1.7	-0.2	2.09	-0.31	-0.24
60-120m	0.52	3.90	-1.75	0.15	2.60	-0.31	-0.22

A MODERN VIEW

BHANDARI ET AL. (2017, QJE): MOTIVATION

- Start from the Ramsey question: how should the government manage taxes and its debt portfolio to smooth tax distortions in incomplete markets?
- The government can trade securities of different maturities, but observed returns need not span fiscal shocks well.
- So the modern question is not just whether maturities can replicate complete markets, but which maturity structure actually hedges fiscal risk in the data.
- They discipline the problem with U.S. data on primary deficits, aggregate consumption, and returns on traded securities.
- The long-run target debt level then solves a static risk-minimization problem:

$$B^* = \arg\min_B \operatorname{Var}\left(R(s)B + g(s)\right).$$

- Intuition: choose the portfolio that minimizes variation in gross financing needs, not the one that exactly spans complete markets.

BHANDARI ET AL. (2017, QJE): THE MATURITY RESULT

- Their Example 4 interprets the multi-asset problem as a maturity-structure problem with two securities: a one-period bill and an infinitely lived console.
- The optimal portfolio is

$$B_t^2 = B^{*,2}, \qquad B_t^1 = B_t - B^{*,2},$$

  where the long-maturity position is fixed and only the short bill moves over time.
- So the government uses long debt for hedging and short debt for shock-by-shock adjustment.
- If total debt rises, average maturity falls, because the extra financing is absorbed by short debt while the console position stays fixed.
- That is a direct maturity-management result, and it echoes the empirical pattern emphasized by Missale and Blanchard (1994).

BHANDARI ET AL. (2017, QJE): WHY IS TARGET DEBT NEAR ZERO?

- In the data-based version, the target debt position is proportional to

$$-\frac{\mathrm{Cov}\left(R_t,\ PV_{\Phi,t}\right)}{\mathrm{Var}\left(R_t\right)}.$$

- So the sign is driven by how returns covary with future fiscal needs.
- In their U.S. calibration, that covariance is slightly positive, so the planner wants a small asset position rather than positive debt.
- The magnitude is close to zero because a large share of return variation is orthogonal to fundamentals: large debt or asset positions buy little hedge and mostly add portfolio risk.
- The same force also helps explain very slow mean reversion: moving debt quickly creates extra return risk.

BHANDARI ET AL. (2017, QJE): QUANTITATIVE U.S. RESULTS

- In their benchmark calibration, the optimal long-run debt target is negative but close to zero: about −6% of GDP.
- The invariant distribution is very dispersed: the standard deviation of debt is about 20% of GDP.
- Taxes move much less than debt: the labor-tax standard deviation is only about 0.5 percentage points.
- Mean reversion is extremely slow: the half-life is about 250 years.
- Interpretation: once returns are measured from U.S. data, debt management looks like slow fiscal-risk management, not exact state-by-state replication.

BEGS (2019): RAMSEY VS. U.S. DEBT PROFILES

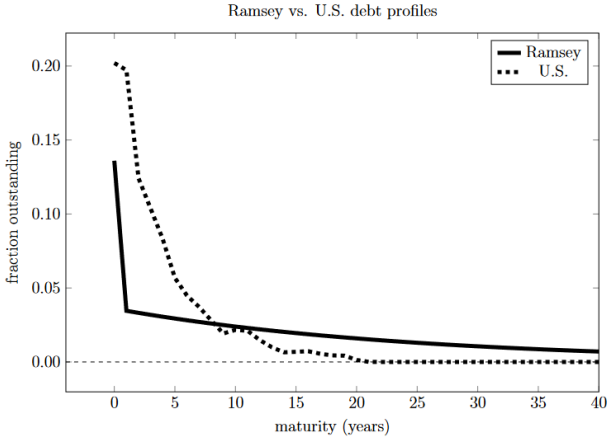


Figure III: Fraction of debt outstanding by maturity (years). For the U.S., we compute outstanding debt by bins of (Macaulay) duration and average for the sample period 1960-2015. The Ramsey debt profile is constructed using a replicating portfolio of zero-coupun bonds of all maturities.

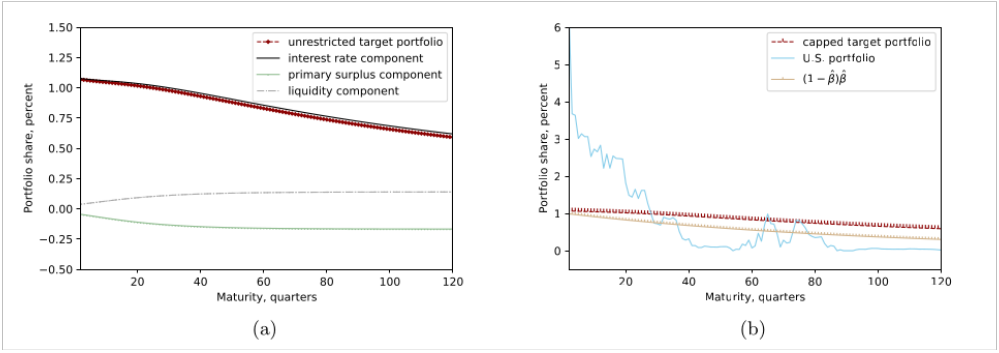
APARISI DE LANNOY ET AL. (2025): U.S. MATURITY RESULTS

- Aparisi de Lannoy, Bhandari, Evans, Golosov, and Sargent (2025, JPE) study a broader public-portfolio problem with richer asset-pricing forces, liquidity benefits, and price effects.
- In their U.S. application, the target maturity portfolio has shares that decline roughly geometrically with maturity.
- That implies a portfolio with longer duration than the current U.S. maturity structure.
- The dominant force is hedging interest-rate risk; primary-surplus, liquidity, and inequality terms are quantitatively small or offset each other.
- Because that target already minimizes rollover risk, the government needs little rebalancing, so the optimal portfolio stays close to the target even when issuance moves prices.

APARISI DE LANNOY ET AL. (2025): WHAT DRIVES THE PORTFOLIO?

- They decompose the target portfolio into interest-rate, primary-surplus, and liquidity components.
- In U.S. data, bond returns co-move only weakly with macro fundamentals, so bonds are poor hedges of primary-surplus and liquidity risk.
- Those two components also offset each other quantitatively.
- That leaves interest-rate / rollover risk as the dominant force: matching debt duration to liability duration gives the roughly geometric maturity weights.
- Because that hedge minimizes expected rollovers, the government needs little rebalancing, so price-impact effects matter less than one might expect.

APARISI DE LANNOY ET AL. (2025): FIGURE 1



SUMMARY

1. Different maturities matter because the market value of inherited noncontingent debt is state dependent.
2. Angeletos' key step is: choose maturities at $t-1$ so that their state-dependent market value at t equals the debt burden / present value of surpluses that contingent debt would have delivered.
3. In the two-state benchmark, this implies holding short assets and issuing long debt.
4. Buera–Nicolini show that exact spanning can require enormous gross positions because return matrices are nearly singular.
5. Bhandari et al. reinterpret maturity management as fiscal-risk minimization: keep the long hedge roughly fixed and use short debt for adjustment; Aparisi de Lannoy et al. show that in U.S. data this is driven mainly by interest-rate / rollover risk.